Probability Distributions

Expected value, and the Bernoulli, binomial, Poisson and normal distributions with their fitting.

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1 Distributions & expectation

Discrete

p.m.f. $p(x) = P(X = x)$; $p(x) \geq 0$, $\sum p(x) = 1$

Continuous

p.d.f. $f(x)$; probability = **area** under the curve; total area = 1

Expected value (discrete)

$$E(X) = \sum x\, p(x)$$

Expected value (continuous)

$$E(X) = \int x f(x)\, dx$$

Expected value = the long-run average of the random variable.

2 Bernoulli

A single trial with two outcomes: success (1), failure (0). Parameter p.

p.m.f.

$$p(x) = p^x (1-p)^{1-x}, \quad x = 0, 1$$

Mean	p	Variance	$p(1-p)$

3 Binomial $B(n, p)$

Number of successes in n **fixed, independent** Bernoulli trials with **constant** success probability p; $q = 1 - p$.

p.m.f.

$$p(x) = \binom{n}{x} p^x q^{n-x}$$

Mean · Variance

$$E(X) = np, \quad Var(X) = npq$$

Fitting: estimate p

$$\hat{p} = \frac{\bar{x}}{n}$$

Variance < mean. Expected frequency = $N \times p(x)$, $x = 0, 1, \ldots, n$.

4 Poisson $P(\lambda)$

Counts of events (0, 1, 2, ...) at a rate; approximates rare binomial successes.

p.m.f.

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Mean = Variance = λ. Fit with $\hat{\lambda} = \bar{x}$.

5 Normal $N(\mu, \sigma^2)$

p.d.f.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- ✓ Continuous, bell-shaped, symmetric about μ; mean = median = mode
- ✓ Skewness 0, kurtosis $\beta_2 = 3$; odd central moments are zero
- ✓ Tails approach the x-axis but never touch it
- ✓ Q.D. ≈ 0.6745σ; M.D. ≈ 0.7979σ

6 Standard normal & areas

Standardise

$$z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

Interval	Area
$\mu \pm \sigma$	68.26%
$\mu \pm 1.96\sigma$	95%
$\mu \pm 2\sigma$	95.44%
$\mu \pm 2.58\sigma$	99%
$\mu \pm 3\sigma$	99.73%

7 Compare the distributions

	Binomial	Poisson	Normal
Type	Discrete	Discrete	Continuous
Parameters	n, p	λ	μ, σ
Mean	np	λ	μ
Variance	npq	λ	σ^2
Agri example	Germinated seeds out of 10	Insects per leaf	Plant height

Z-table: area to the left of z. Above z: $1-$ table value. Between two z's: subtract.

! Common mistakes

- × Binomial needs a **fixed** n and **constant** p.
- × $P(X = x) = 0$ for continuous variables: use intervals.
- × Forgetting $1-$ table value for upper-tail areas.

? Fill in the blanks

1. A binomial variable with n = 10 and p = 0.3 has mean _____ and variance _____. [3; 2.1]
2. If 80% of seeds germinate, the expected number germinating out of 20 is _____. [16]
3. A Poisson variable with variance 4 has mean _____. [4]
4. Plant height ~ N(100, 10^2): a plant of 120 cm has z = _____. [2]
5. About _____ % of a normal population lies between μ − 2σ and μ + 2σ. [95.44]
6. In a binomial distribution the variance is always _____ than the mean. [less]

Source: *Textbook of Agricultural Statistics*, Ch. 11 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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